HKU Faculty of Engineering

Tam Wing Fan Innovation Wing

3D Motion Capture System

User Guide

by motion capture team

VICON NEXUS

Ver. 0.10

Last Update: 02 April 2026

About This Manual

Manual Name	3D Motion Capture System - User Guide
Manual Version	0.10
First Version	0.10

About This Equipment

Equipment name	3D Optical Motion Capture System
Location	Event Hall
Software	VICON NEXUS 2.16
Accessories	Active Wand Reflective markers (hard) X base markers (soft) Suit Dummy

About Us

If you have any questions, feel free to contact with our motion capture team.

Email: ehank2142@connect.hku.hk (Hank Xu)

Table Of Contents

1. CALIBRATION 4

1.1 Start the system 4

1.2 Video calibration setup 4

1.3 Mask camera 8

1.4 Calibrate cameras 9

1.5 Set floor origin/plane 11

1.6 Save / Load calibration file 15

2. CAPTURE DATA 16

2.1 Database management 16

2.2 Create a subject 18

2.3 Marker placement 20

2.4 Start Capturing 23

2.4.1 Static Trial 23

2.4.2 Dynamic Trial 27

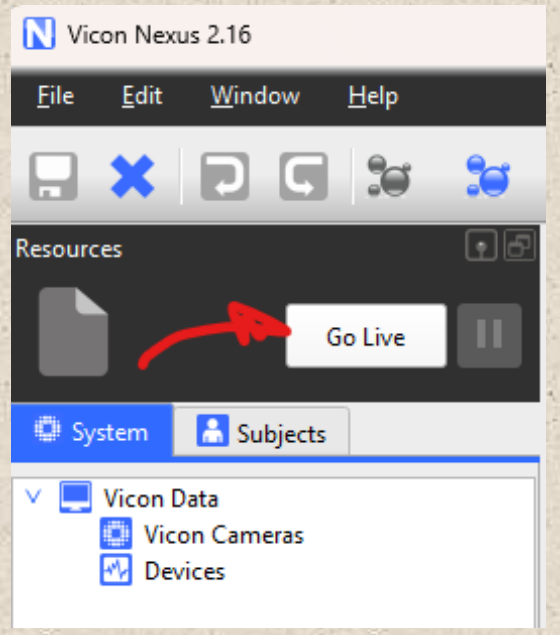
3. VIEW AND EXPORT 31

1. Calibration

1.1 Start the system

Open the Vicon Nexus 2.16 software

Click “Go Live” to start the motion capture system.

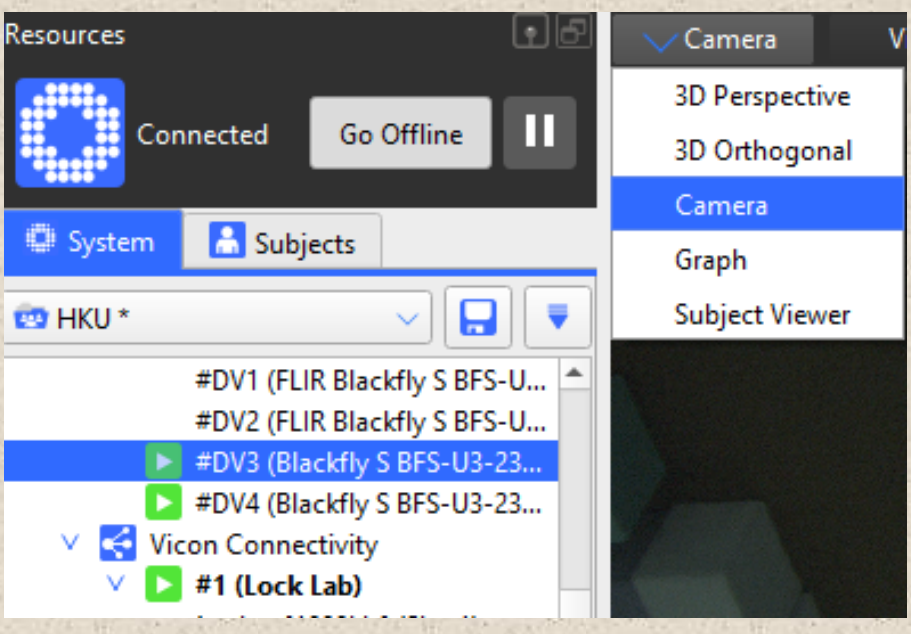


1.2 Video Calibration Setup

Adjust your view:

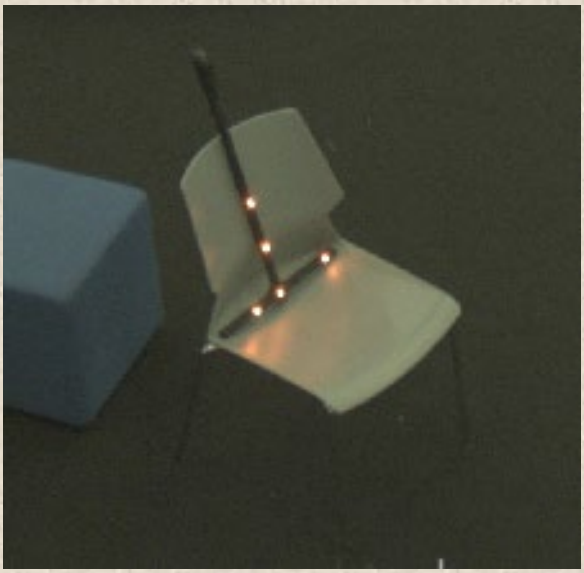
1 Choose Camera perspective.

2 Select all video cameras. (Ctrl + left click)

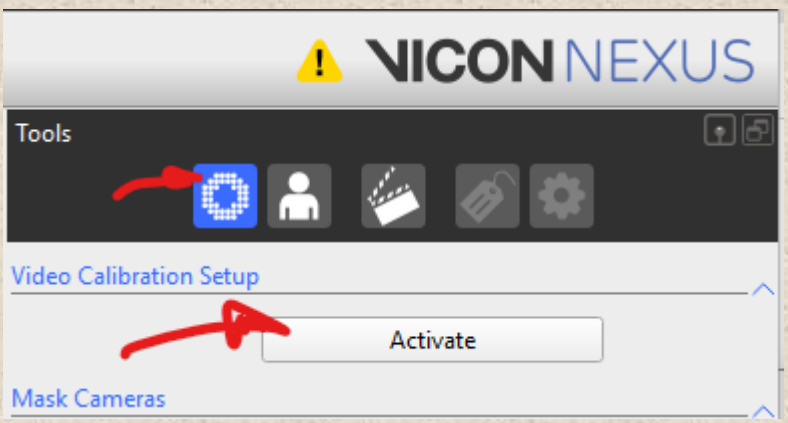


Switch on the Active Wand v2 and put it on a chair in the volume.

Make sure that from all video cameras you can see the wand.



Go to calibration panel “Video Calibration Setup” section and click Activate.



Zoom in to see the quality.

(Right Click Hold + move mouse)

bad example good example

A successful video calibration shall have a cross on all 5 points.

Otherwise, try to adjust

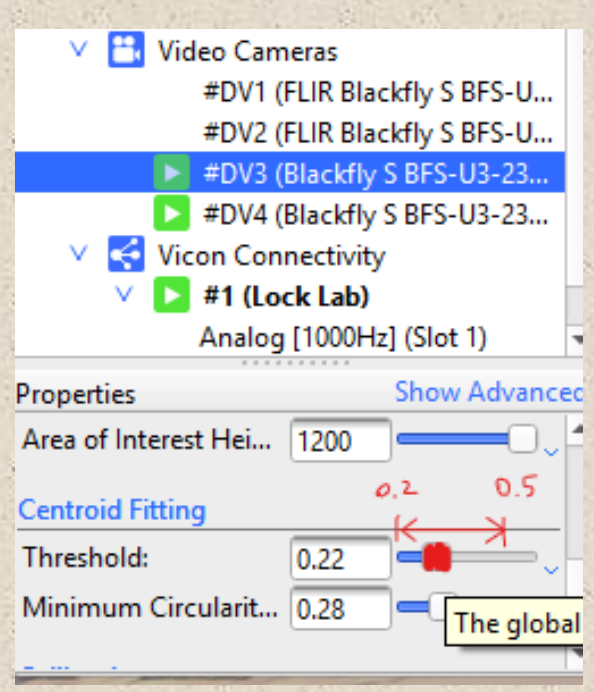
(1) the position of the chair, OR

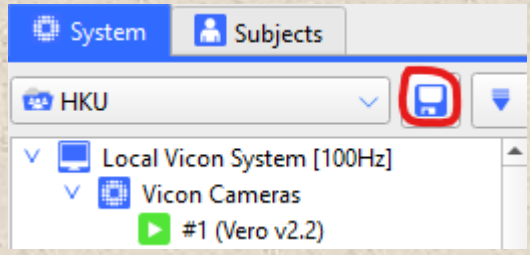
(2) the threshold of the video camera (normal range 0.2-0.5)

to obtain a successful video camera calibration.

Q: How to change threshold?

A:





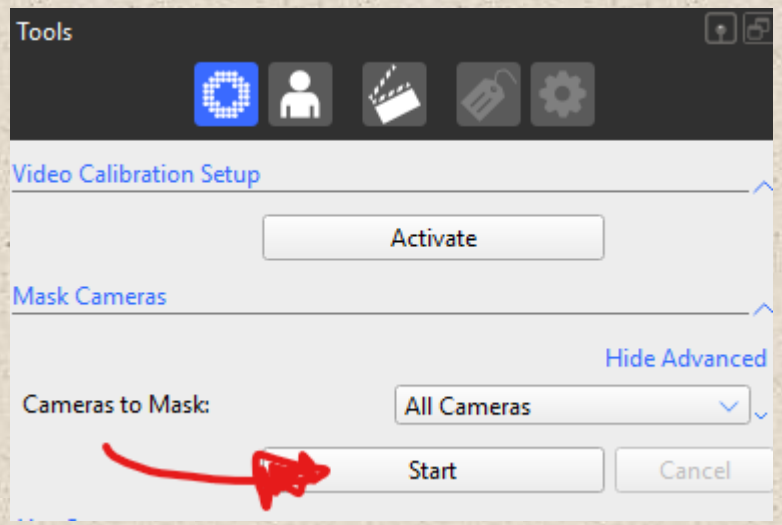
If you adjust the threshold, remember to save.

Upon finishing calibration, click “deactivate”.

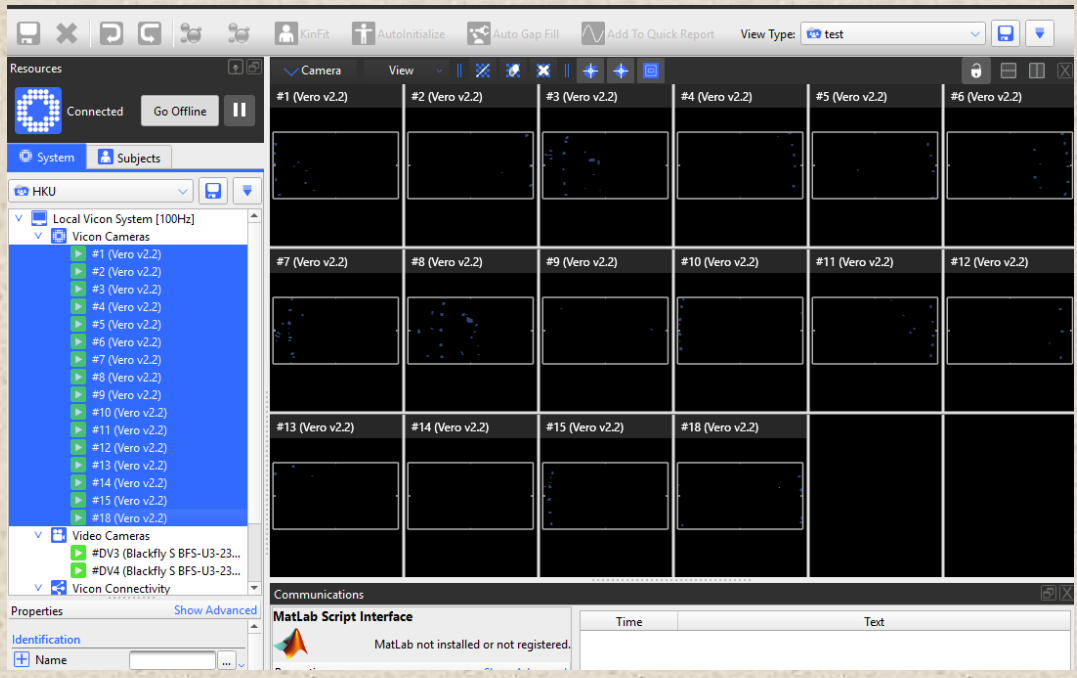
1.3 Mask Cameras

As there are some unwanted reflections in the volume (e.g., reflections from camera metal), we need to mask these reflections.

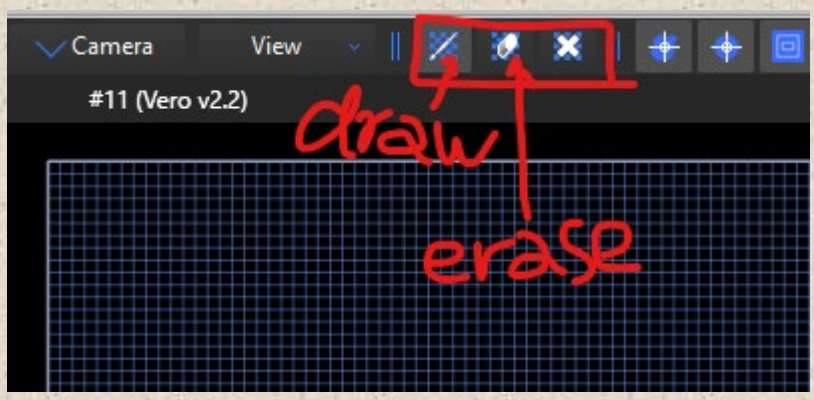
Please first clean the volume. Remove all possible non-fixed items that may cause reflections.



You can choose to mask all cameras or selected cameras.



Wait a moment and check the progress. Remember to click stop once finished.

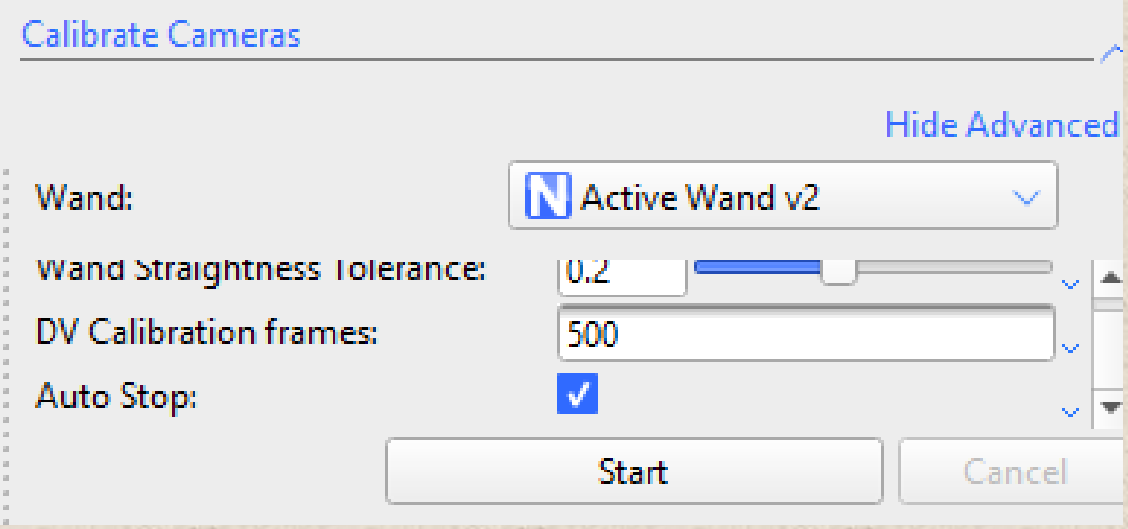


You can also you the tool at the top to mask manually

1.4 Calibrate cameras

Click start on the “calibrate cameras” section.

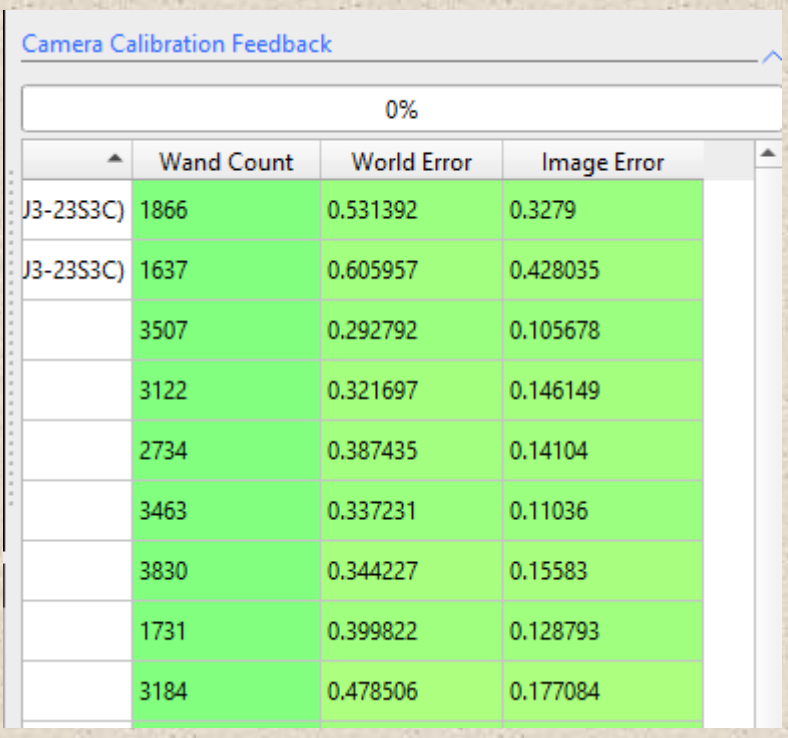
In the capture volume, wave the calibration wand throughout the area, ensuring that the markers on the wand are visible to the cameras.



The light on the camera begins to flash faster when it captures more wanted data. It will turn green once it finishes

calibration. Stop waving when all cameras turn green.

Then wait for software to process.

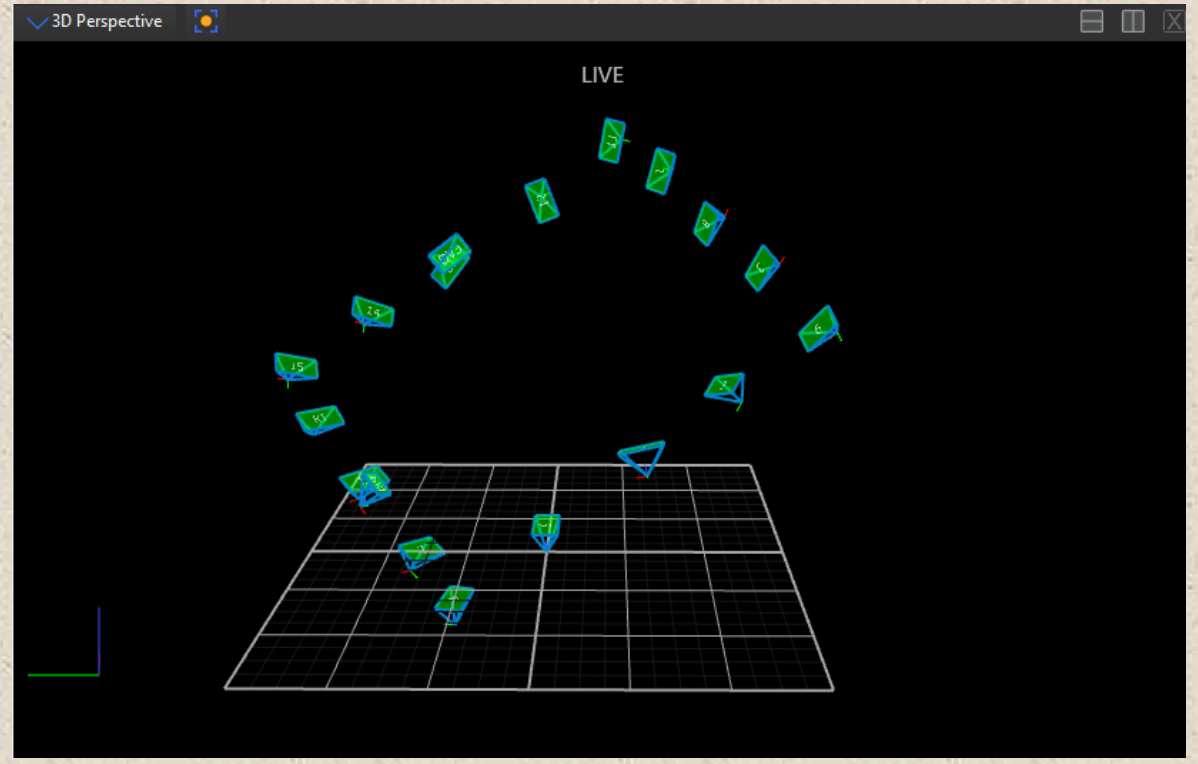


	Wand Count	World Error	Image Error
J3-23S3C)	1866	0.531392	0.3279
J3-23S3C)	1637	0.605957	0.428035
	3507	0.292792	0.105678
	3122	0.321697	0.146149
	2734	0.387435	0.14104
	3463	0.337231	0.11036
	3830	0.344227	0.15583
	1731	0.399822	0.128793
	3184	0.478506	0.177084

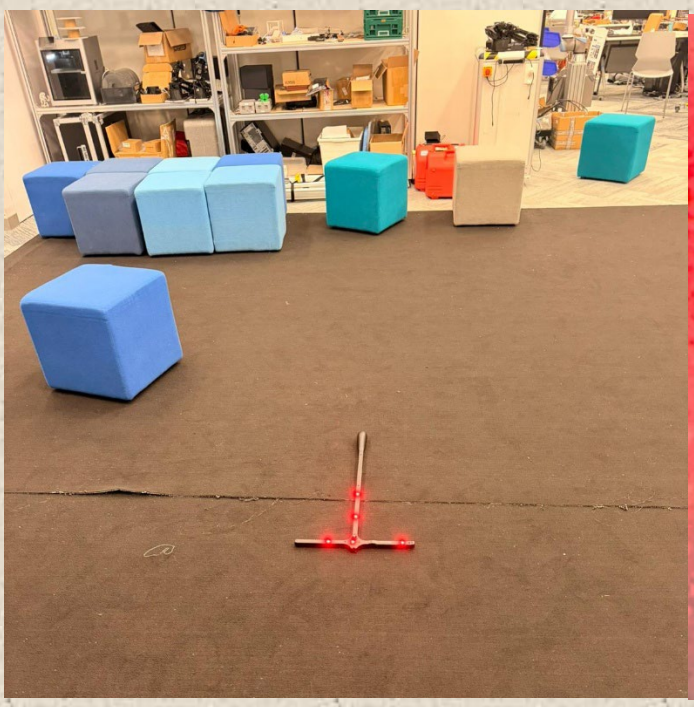
Check calibration quality here

1.5 Set floor origin/Plane

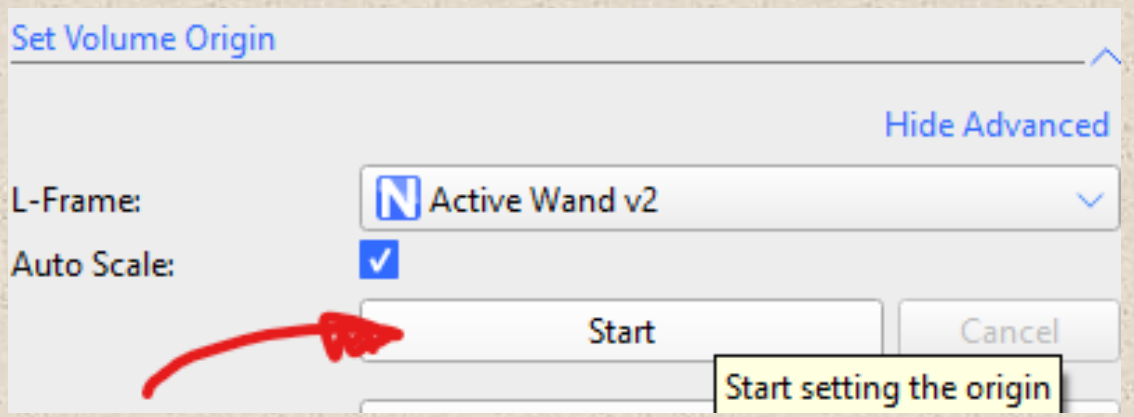
Set floor origin

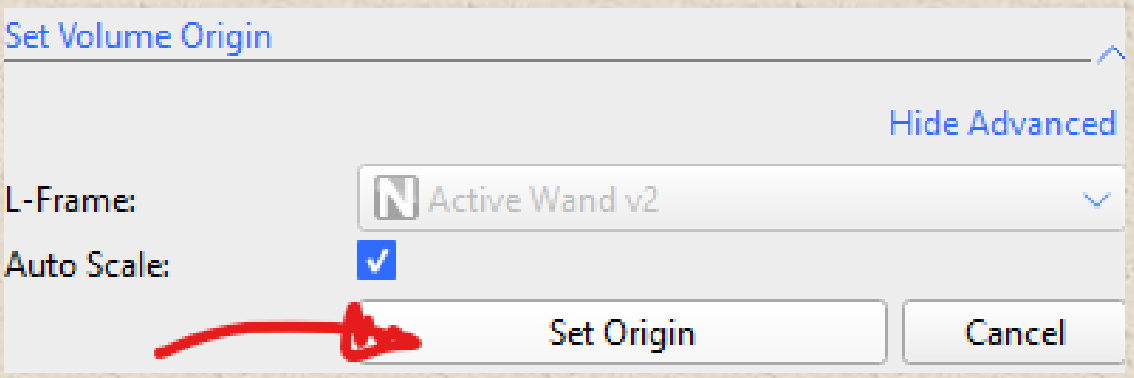


We need to set floor origin.



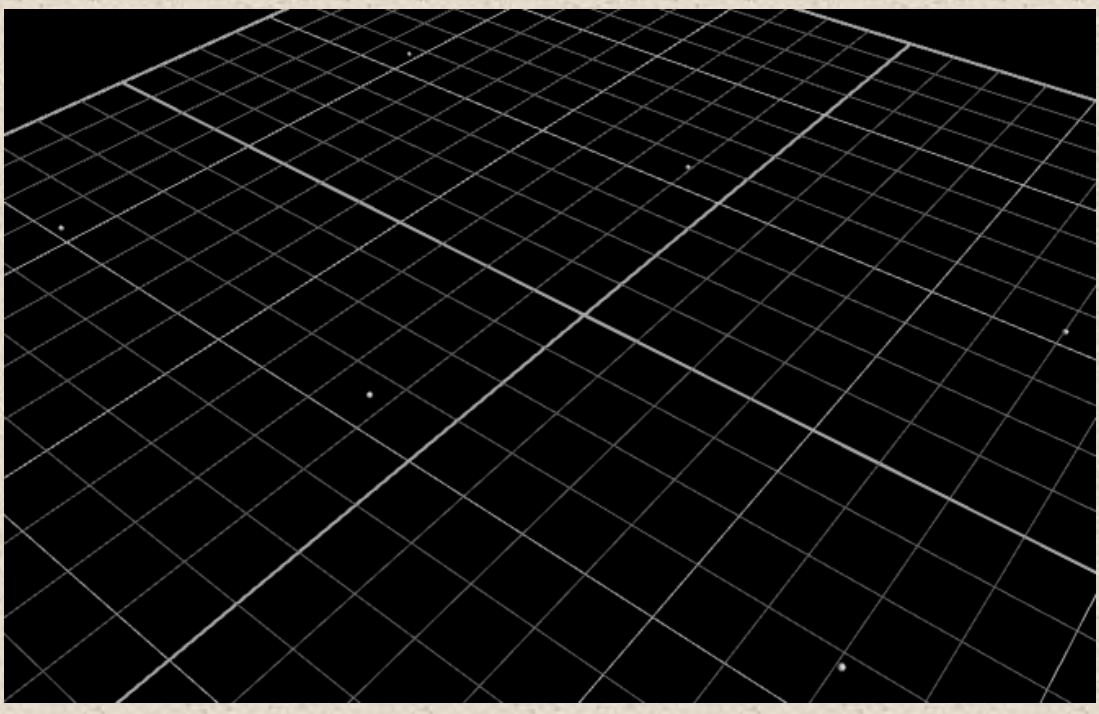
Put the wand at the centre of the volume. Adjust the tilt of the wand to make the mercury ball stay stably in the red circle.

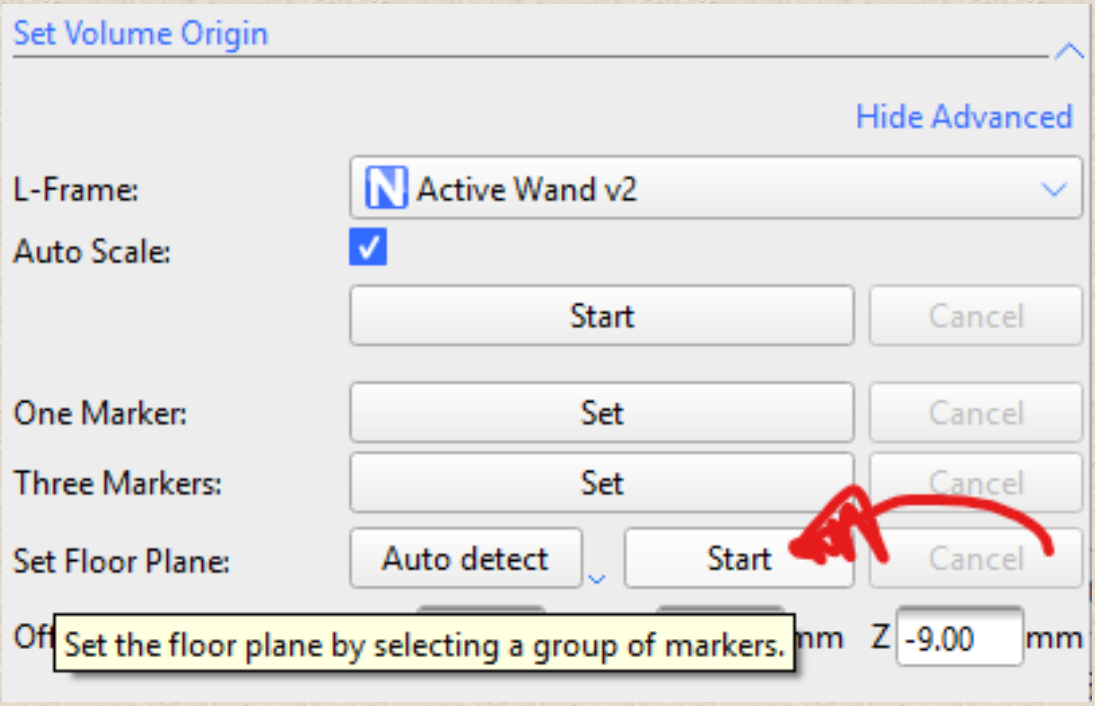




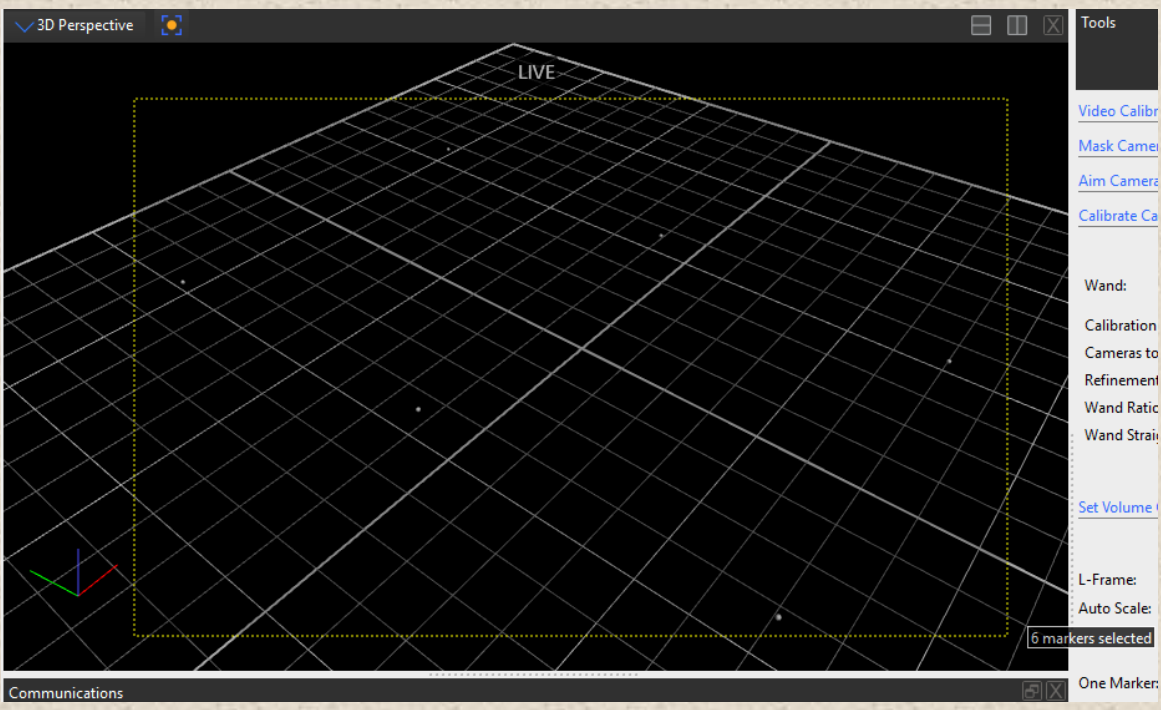
Set floor Plane

Put 6 markers on the floor. (Please remove the wand and turn it off)

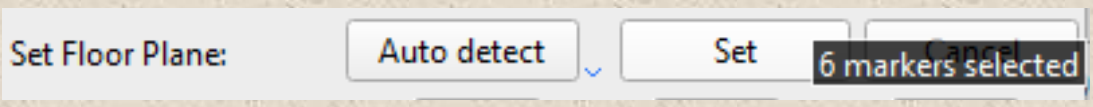



Set Volume Origin
Hide Advanced
L-Frame:
Active Wand v2
Auto Scale:
Start
Cancel
One Marker:
Set
Cancel
Three Markers:
Set
Cancel
Set Floor Plane:
Auto detect
Start
Cancel
Set the floor plane by selecting a group of markers.
Z
-9.00
mm

Click start



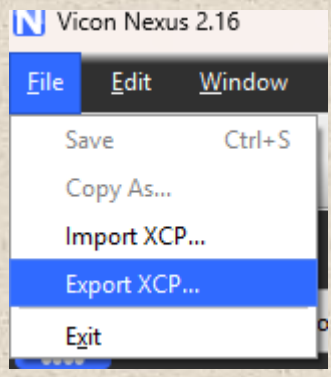
Select 6 markers (alt + drag)

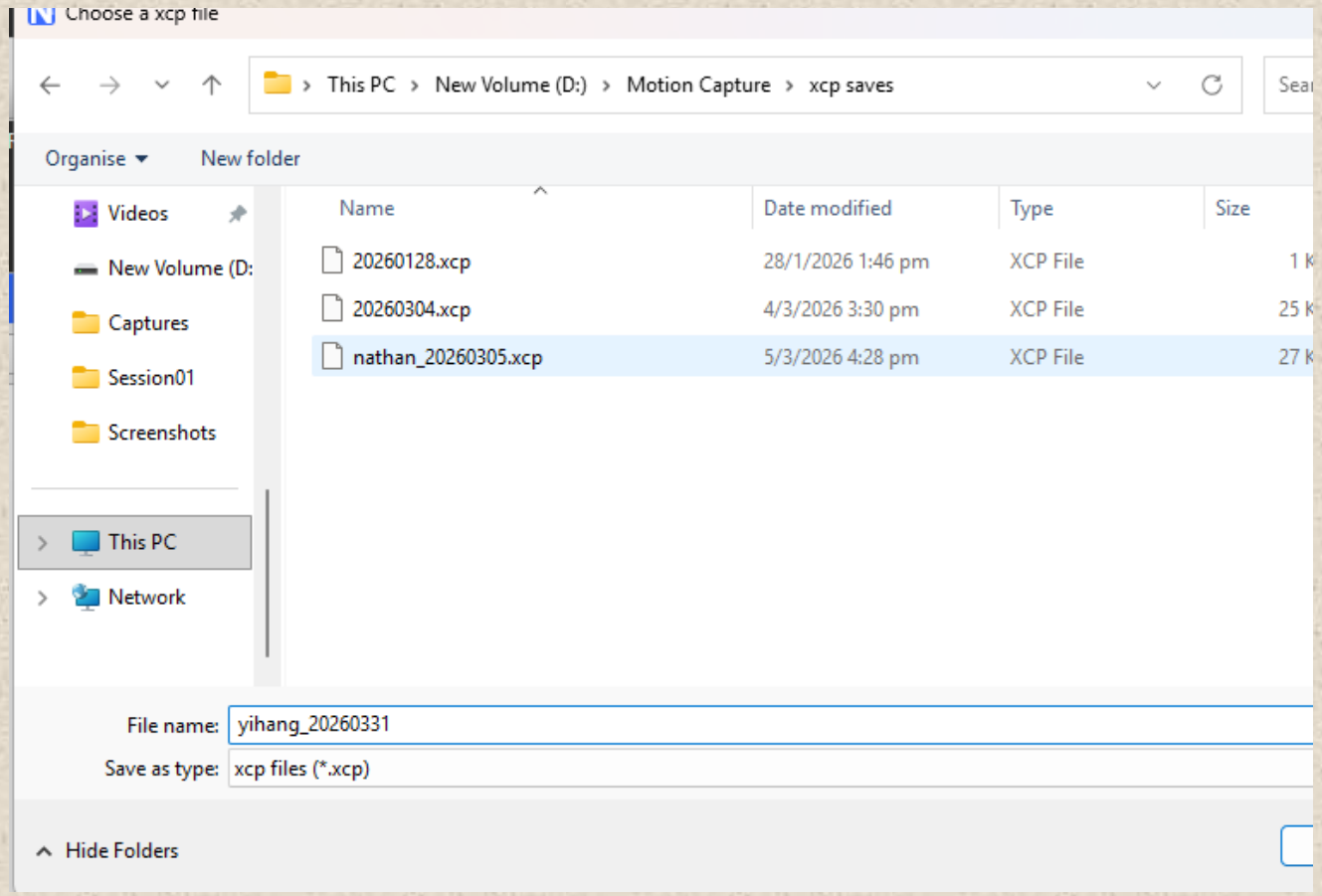


Click set

1.6 Save / Load calibration files

A good habit is to save the calibration file, once you successfully finish calibration.



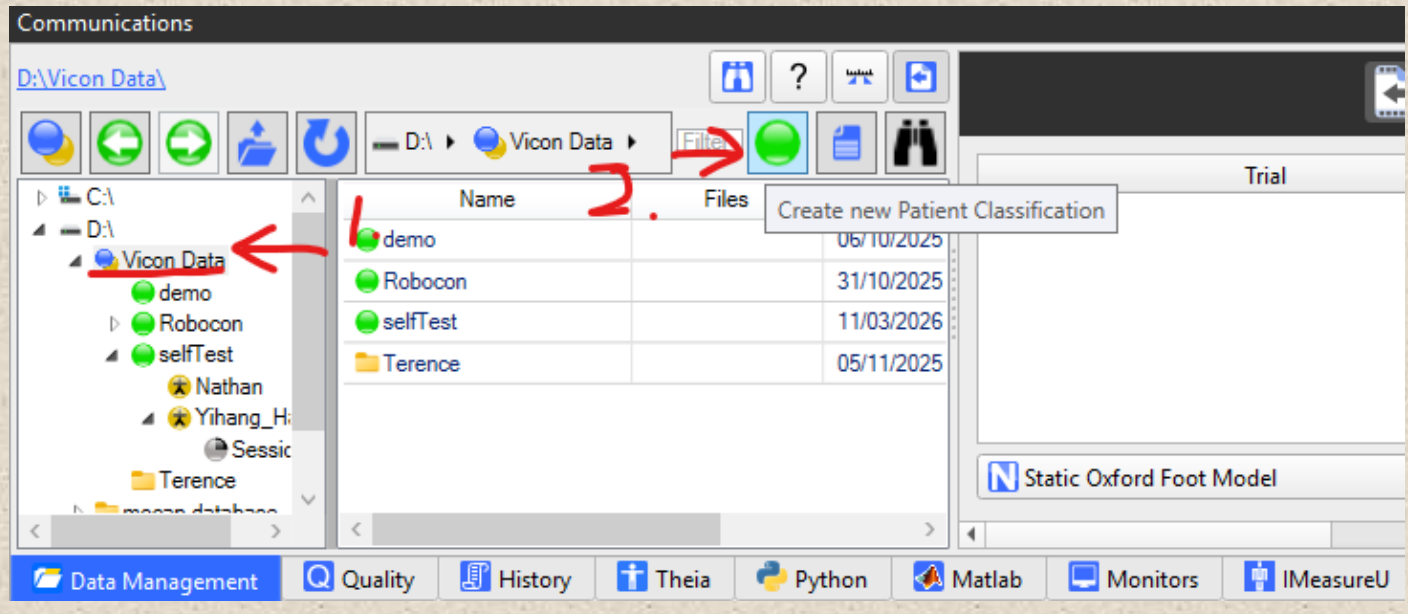


D:\Motion Capture\xcp saves

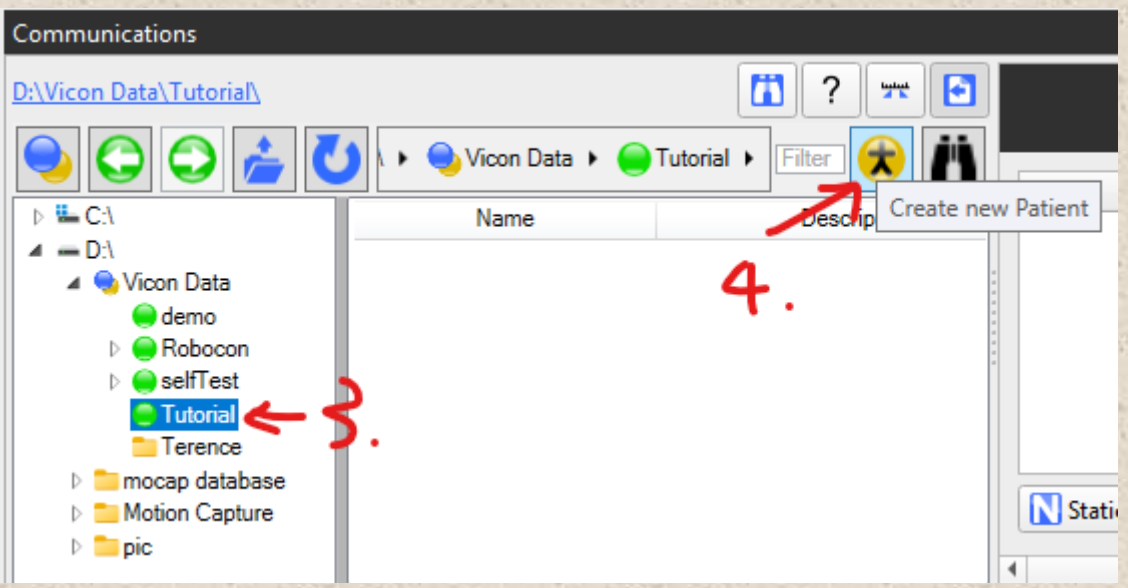
2. Capture Data

2.1 Database management

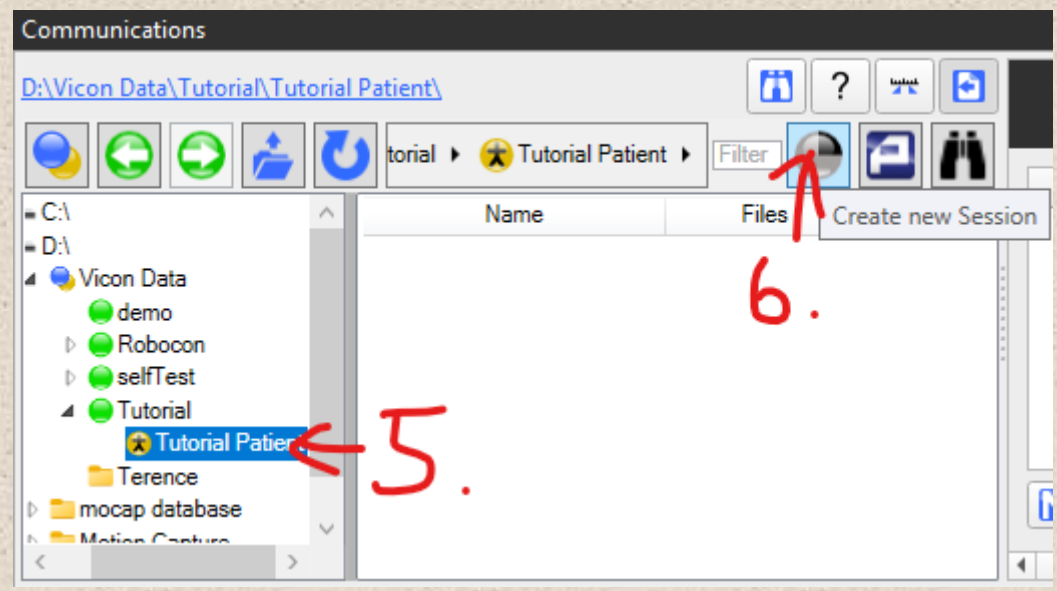
To manage subject and session files orderly, please, create your own database.



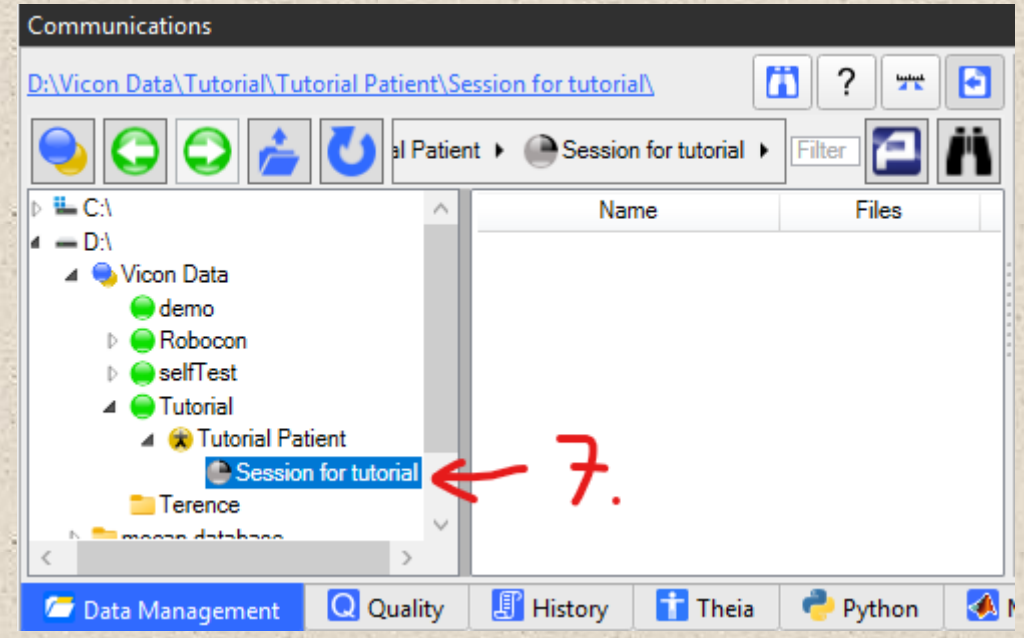
Create your patient classification



Create a new patient



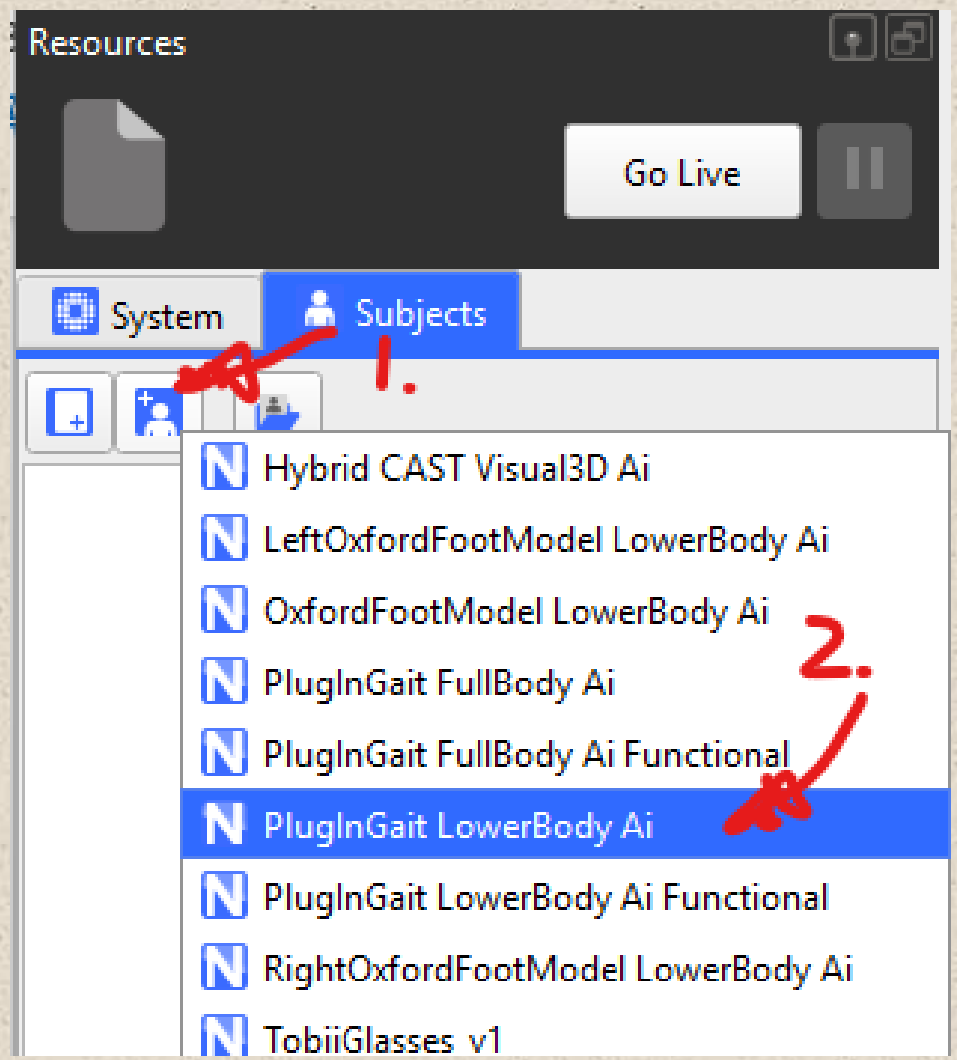
Create a new session



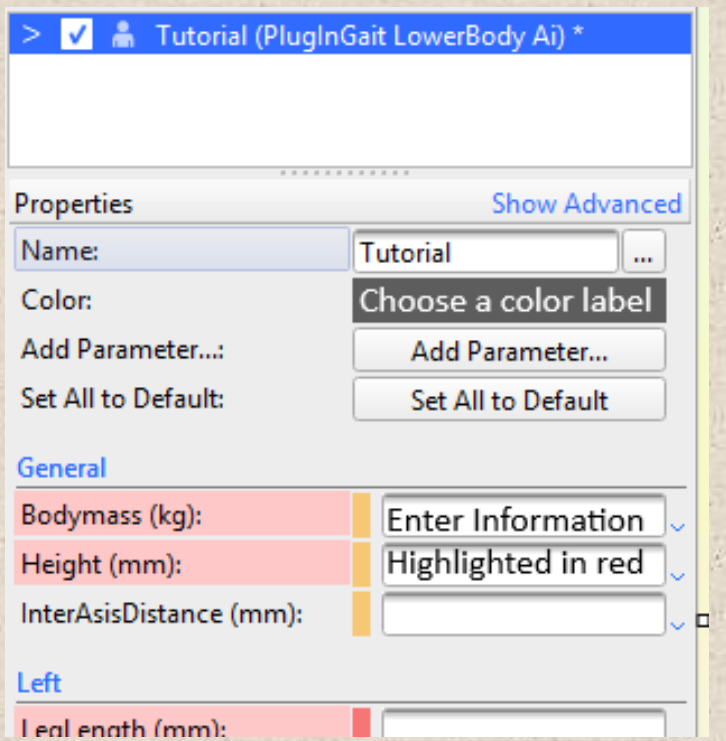
Select this session

2.2 Create a subject

You can only create a subject once you select a session.

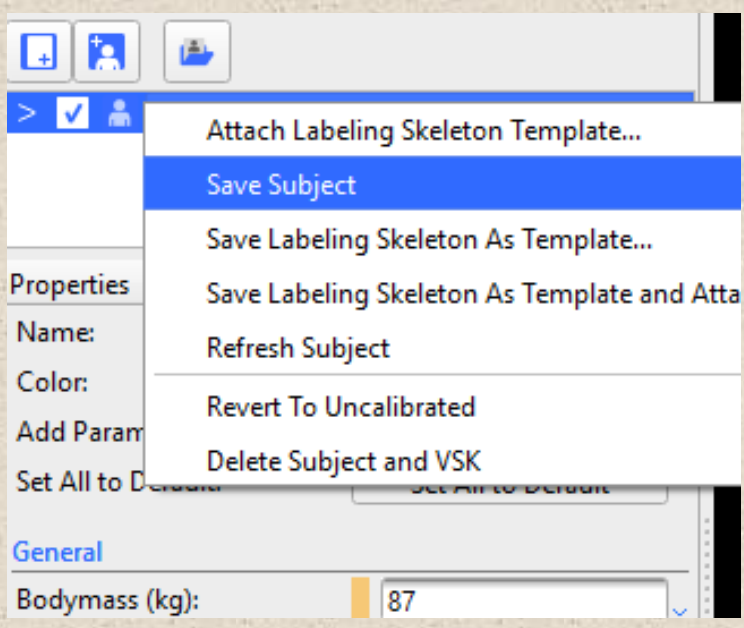


Select a VSK (Vicon Skeleton) that you want to use. This manual will take PluginGait LowerBody Ai as an example.



Click the subject you just created.

Enter the required information highlighted in pink color.

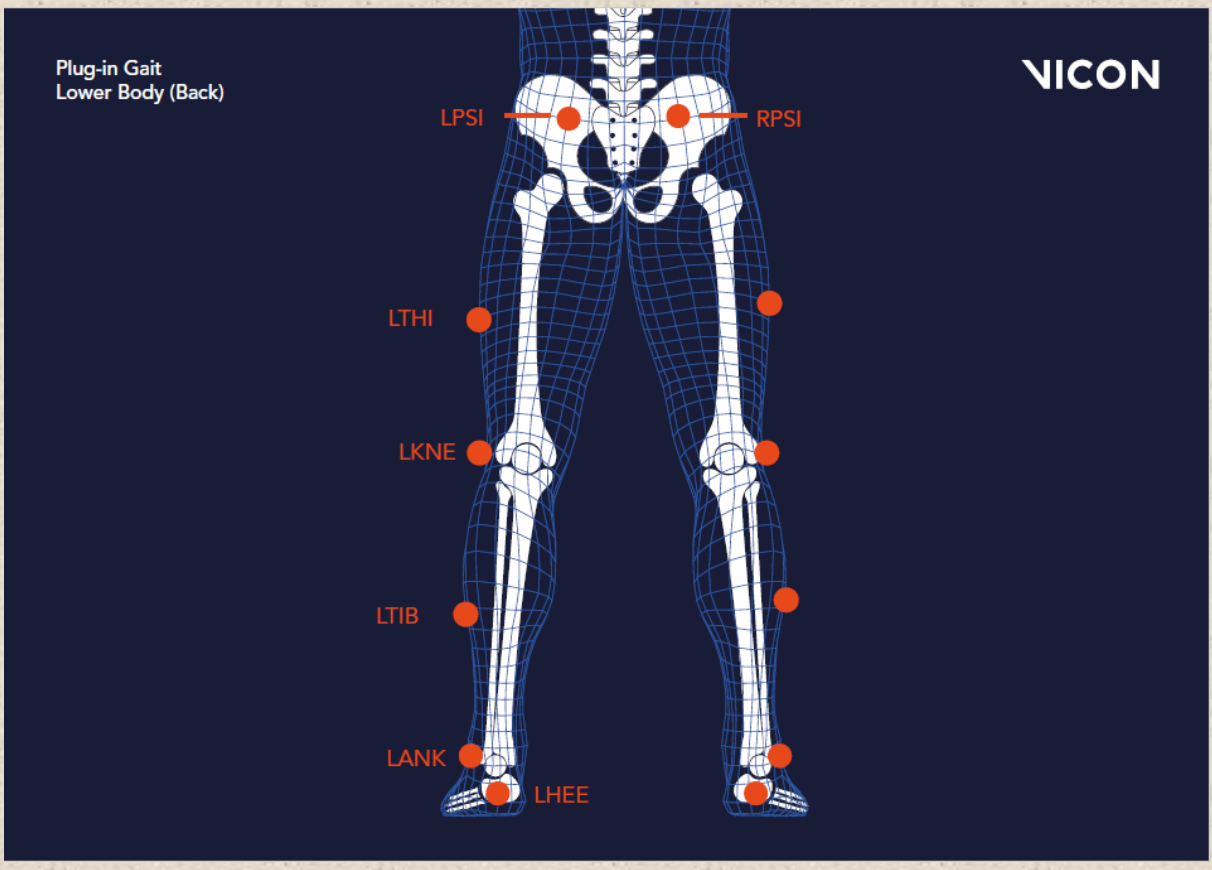


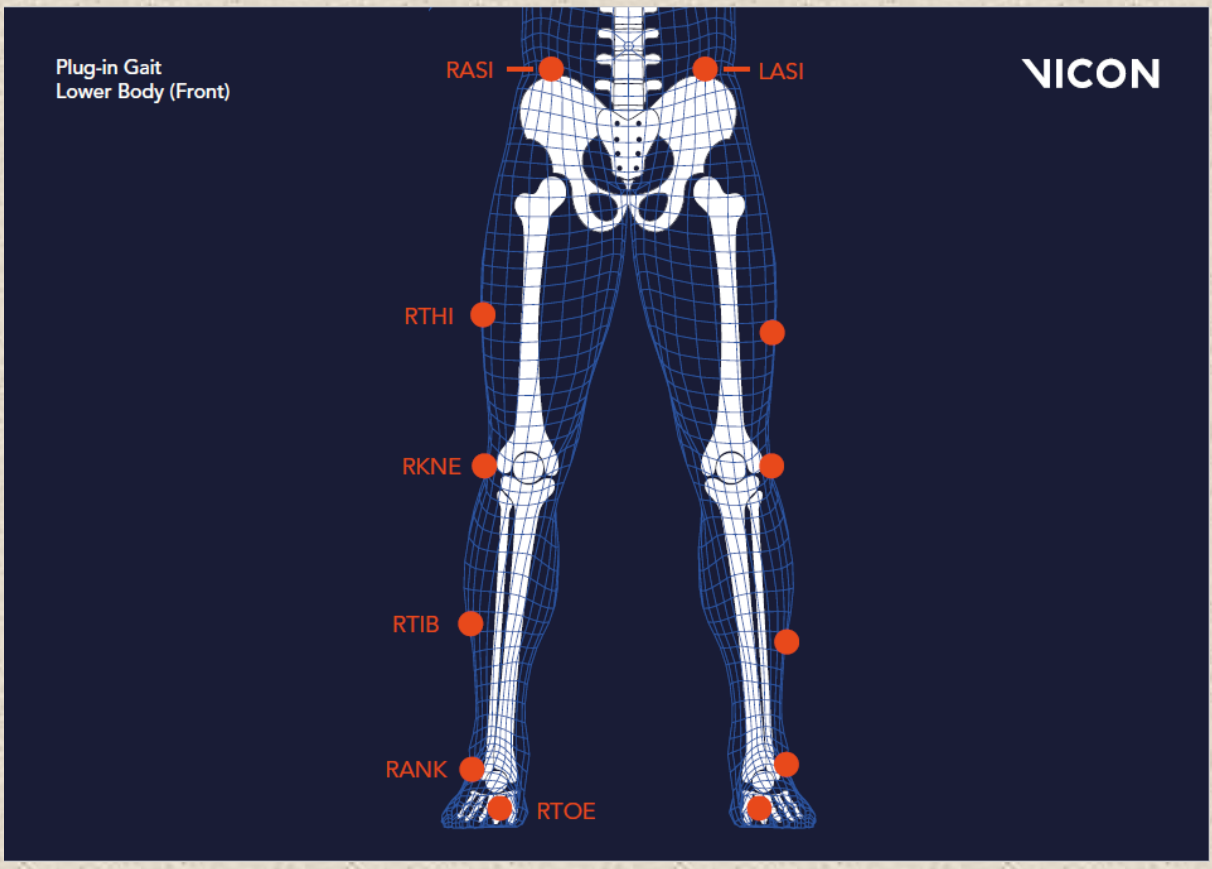
Remember to save subject.

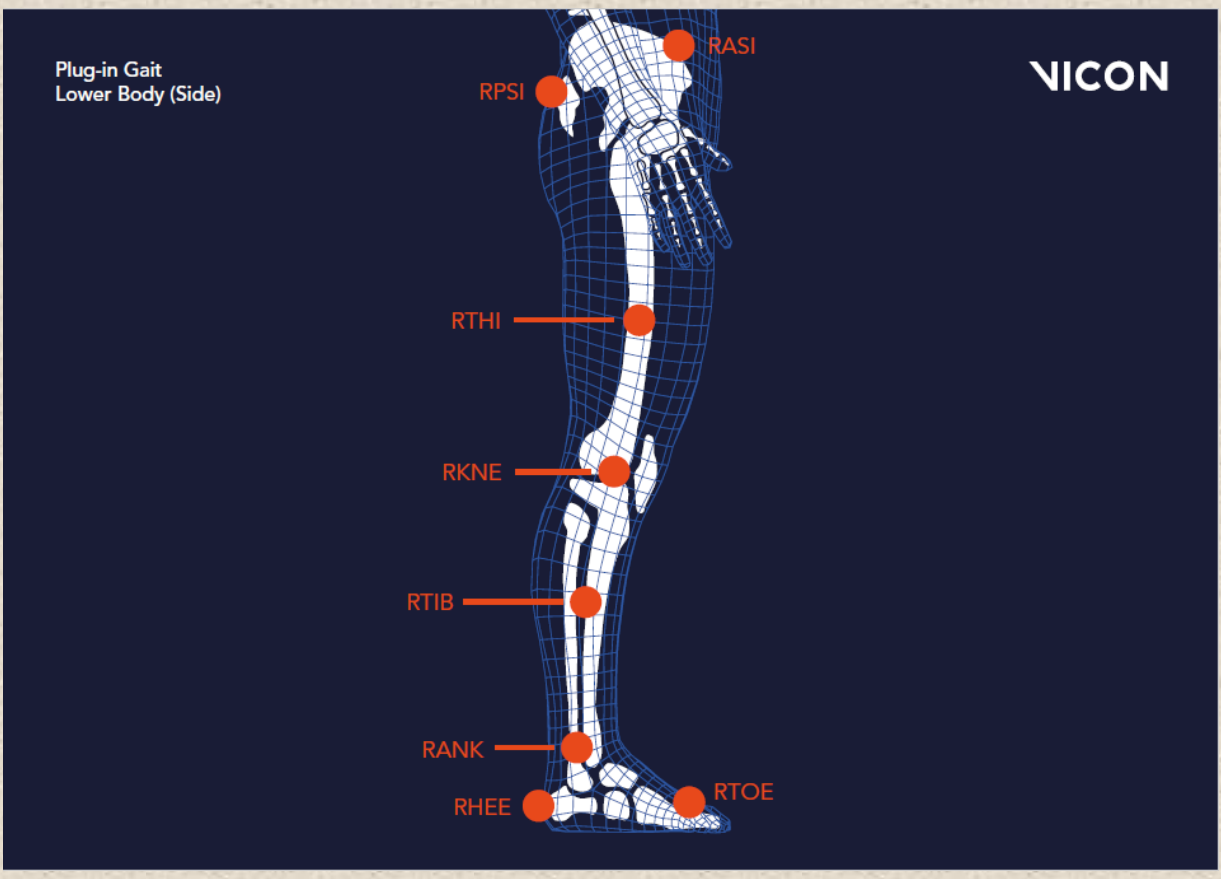
2.3 Marker Placement

[Plug-in Gait models and templates](https://help.vicon.com/space/Nexus216/11614342/Plug-in+Gait+models+and+templates)

https://help.vicon.com/space/Nexus216/11614342/Plug-in+Gait+models+and+templates







An example of lower body plug in gait marker suit (16 markers).

You are advised to wear the suit first on your own, or on him.

For joint markers, please stick the markers exactly on your joint.

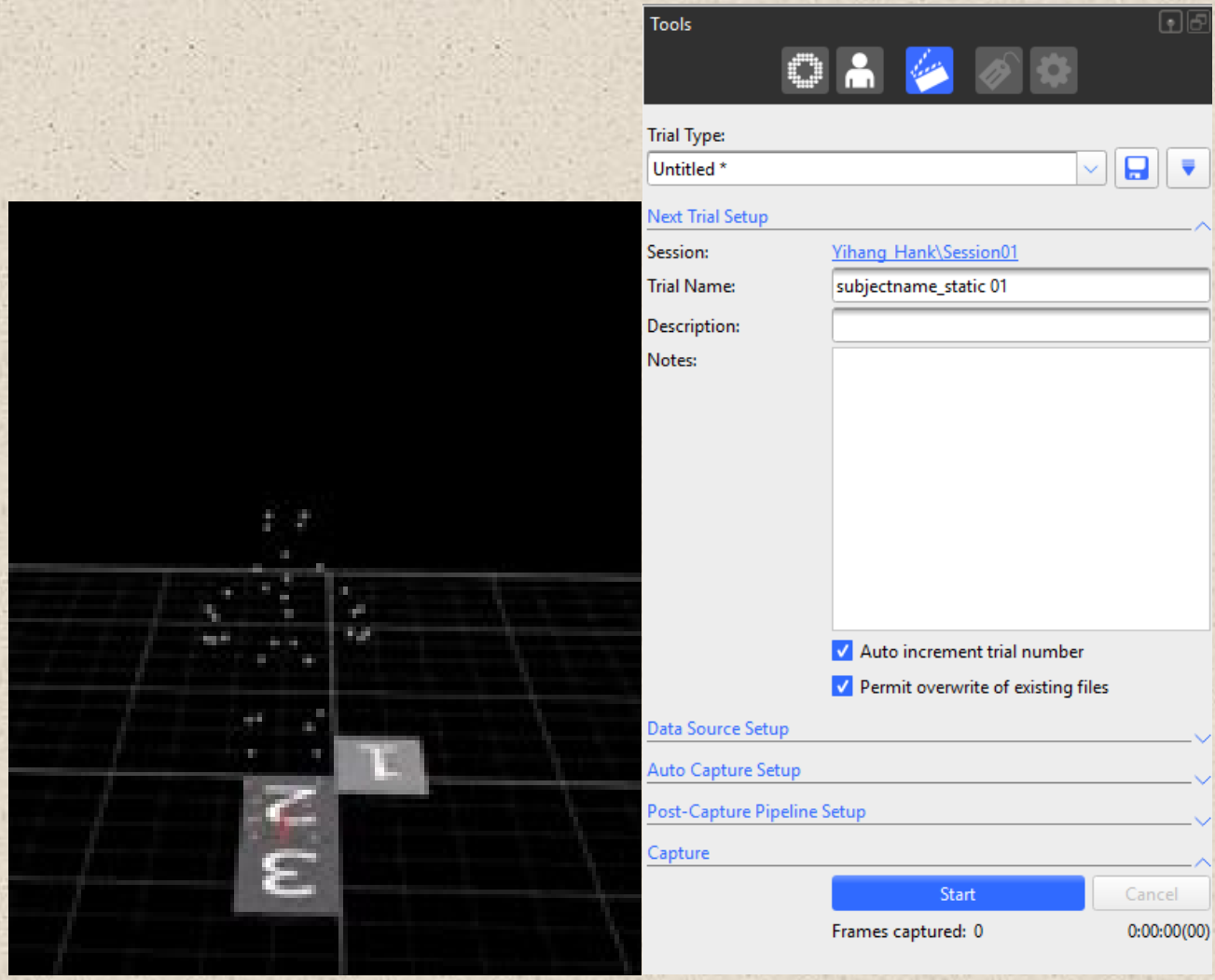
2.4 Start Capturing

2.4.1 Static Trial

Static Trial - *Capturing*

Go Live first.

Wear your marker suit and walk into the volume and stand still, in Motorbike pose.



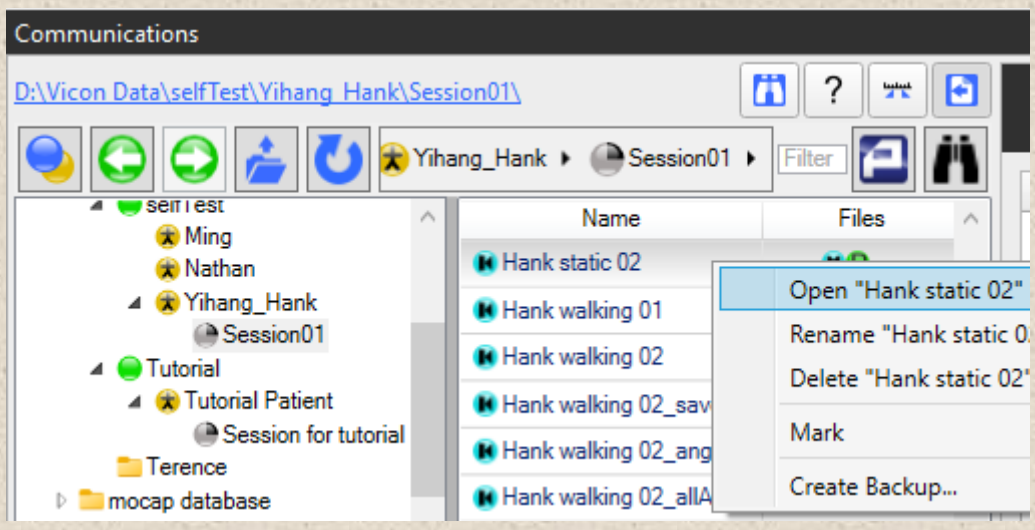
Go to pipeline tool, give the trial a name “xxx static 00”.

Then click start.

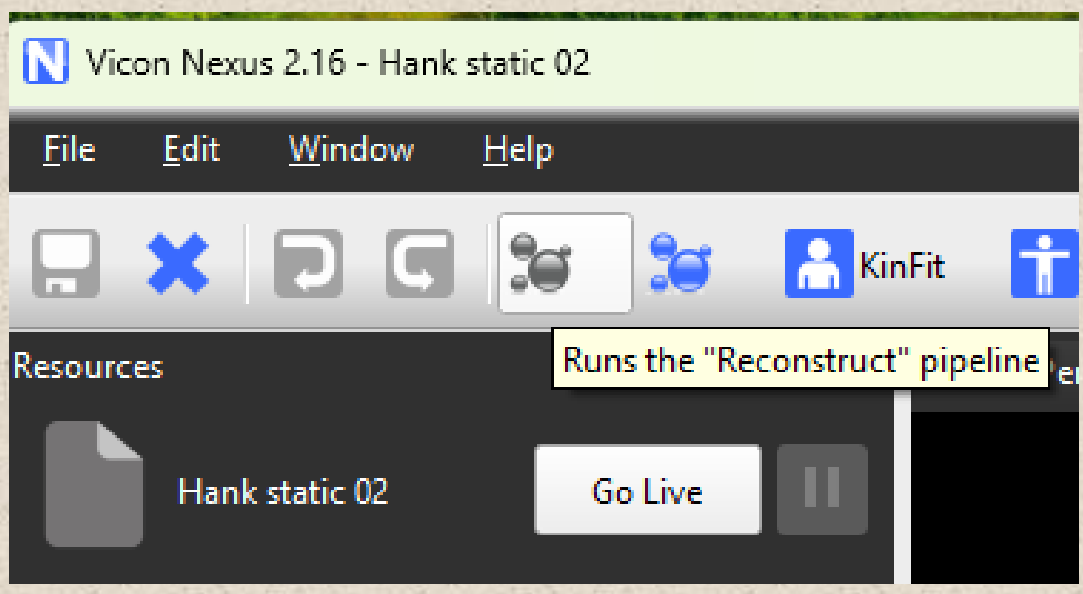
Hold on for about 5 seconds, then stop and relax.

Static Trial - *Processing*

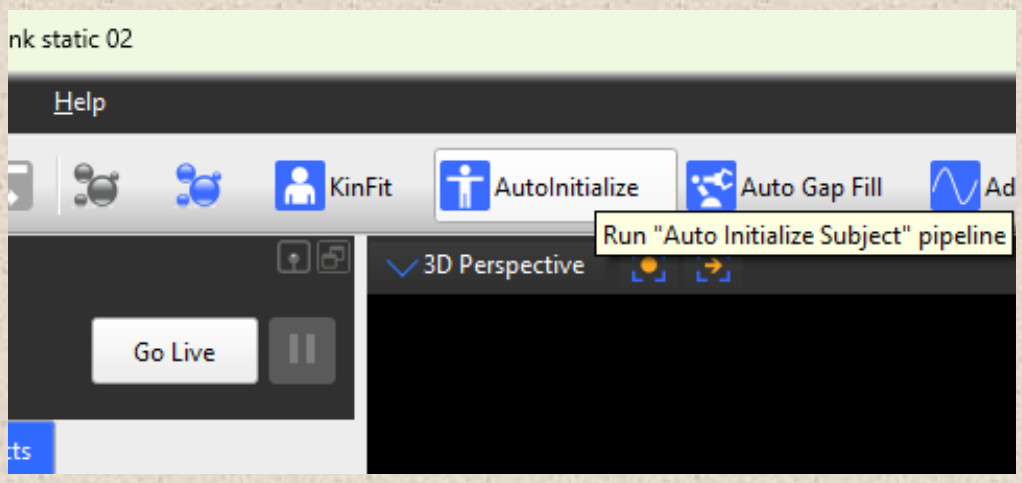
1. Select and open the static trial.

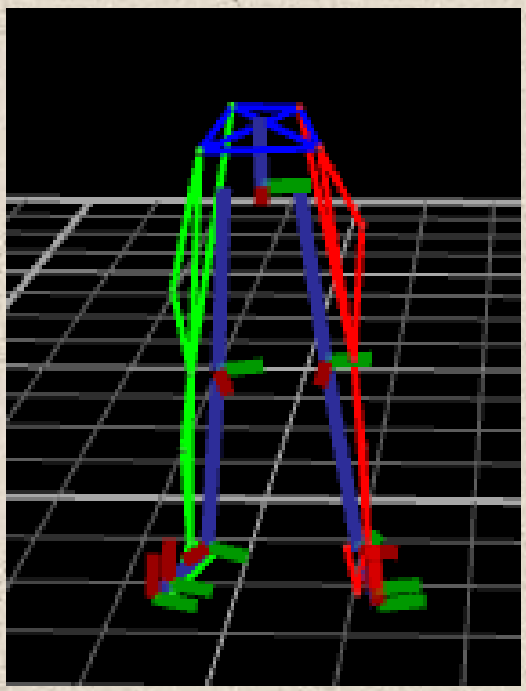


2. Run the reconstruct pipeline. Markers will appear.

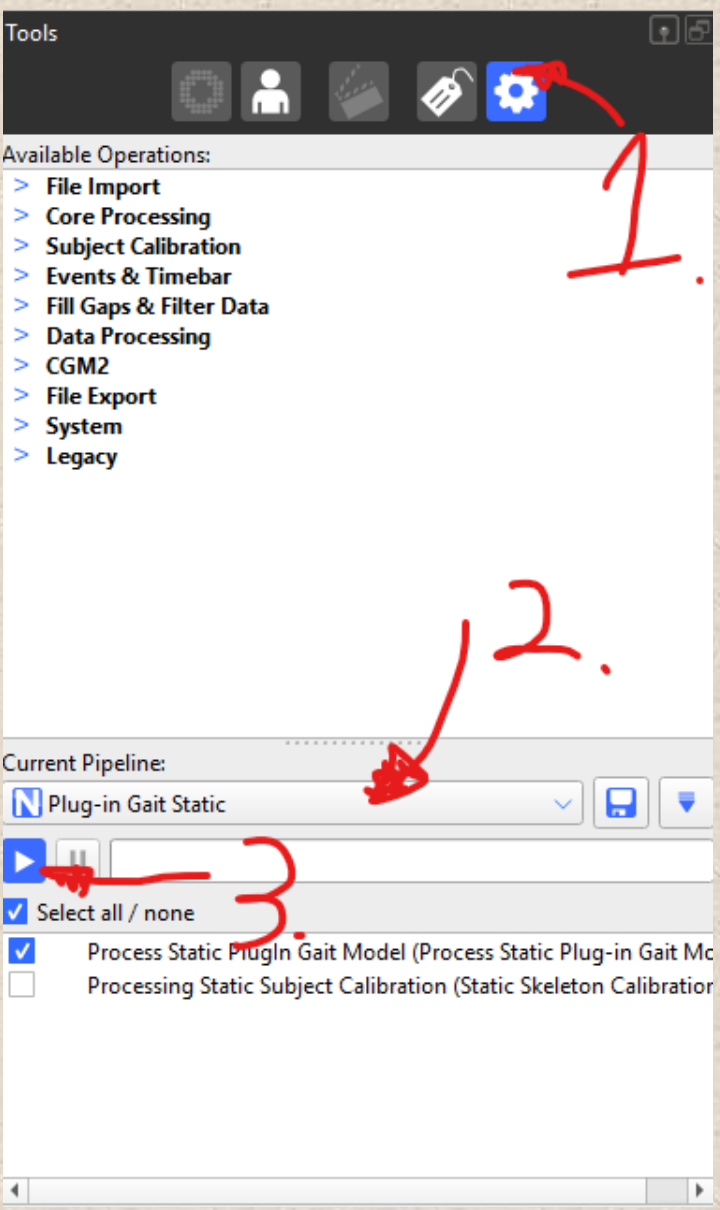


3. Run the Auto Initialize Subject. Skeleton will appear.





4. Go to operations tool. Select Plug-in Gait Static and start processing.


Tools
Available Operations:
File Import
Core Processing
Subject Calibration
Events & Timebar
Fill Gaps & Filter Data
Data Processing
CGM2
File Export
System
Legacy
1.
2.
Current Pipeline:
Plug-in Gait Static
3.
Select all / none
Process Static PlugIn Gait Model (Process Static Plug-in Gait Mc
Processing Static Subject Calibration (Static Skeleton Calibratior

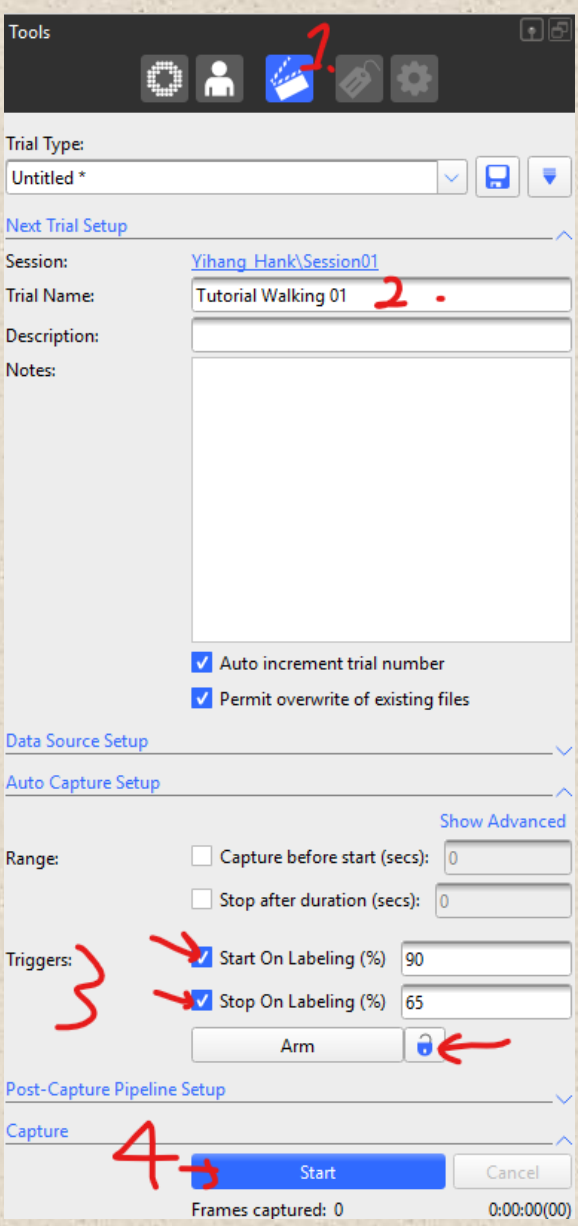
2.4.2 Dynamic Trial

Dynamic Trial - *Capturing*

Go Live.

Go to Pipeline tool. Give the trial a name “xxx [movement] 00”

Arm and lock the trigger, then start.

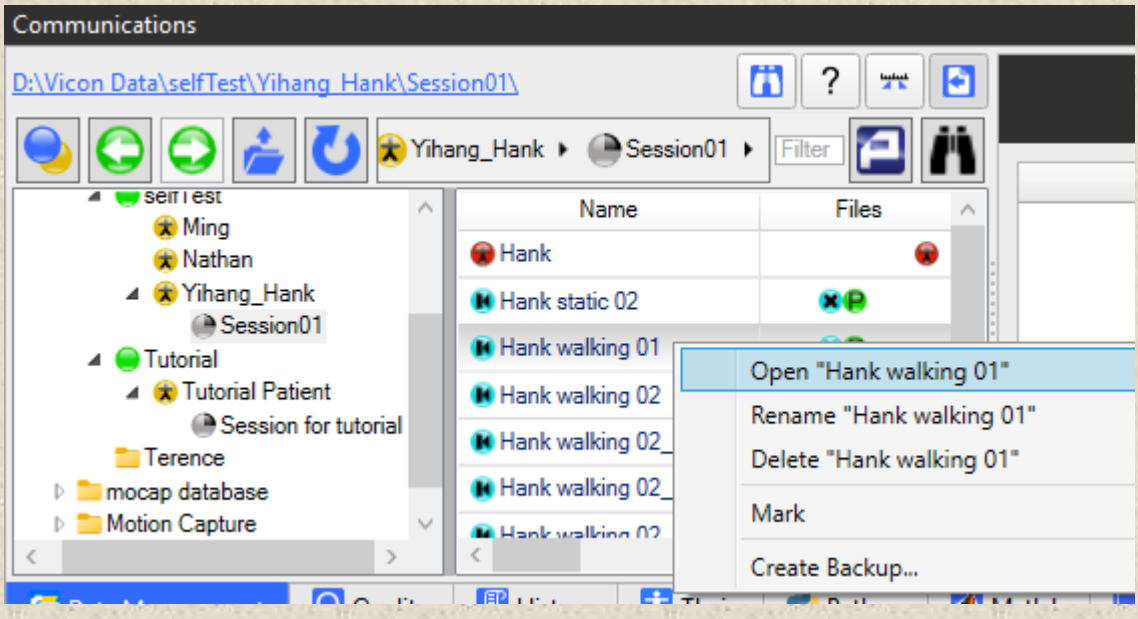


Example: The subject walks into volume. Since the trigger is on, the software would start/ stop capturing only xx% markers are in volume.

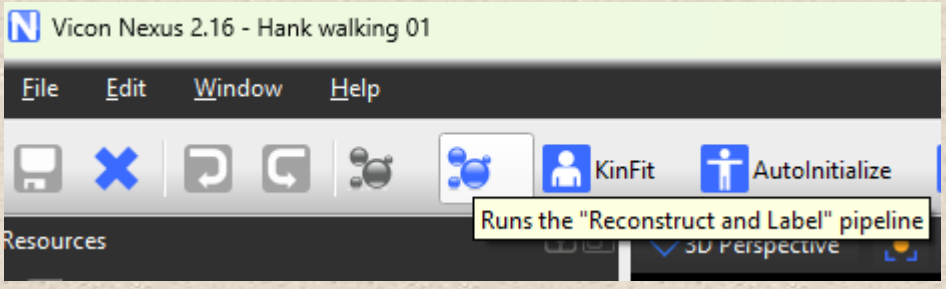
Manually click stop when you want to stop.

Dynamic Trial - *Processing*

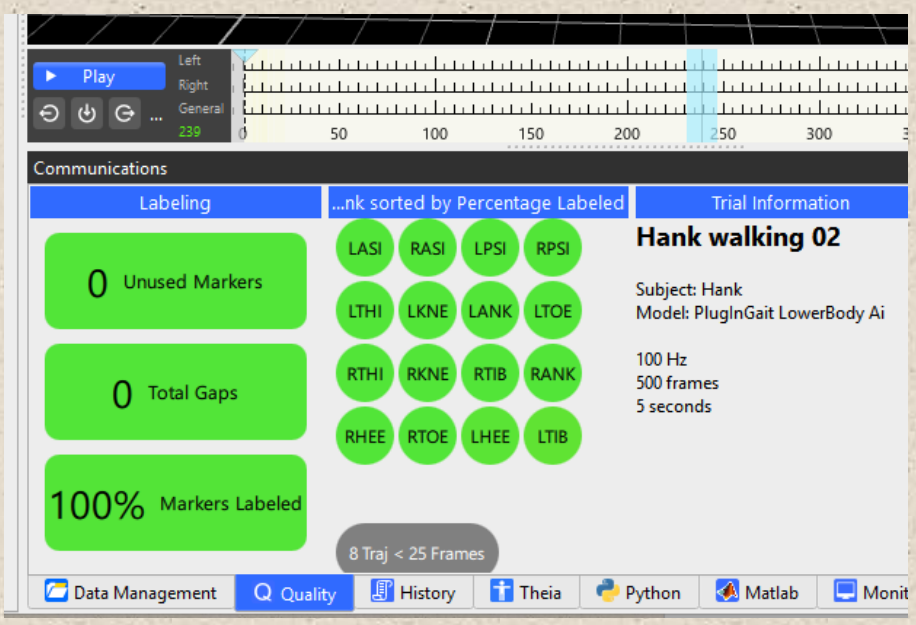
1. Open the captured file.



2. Run the “Reconstruct and Label” pipeline



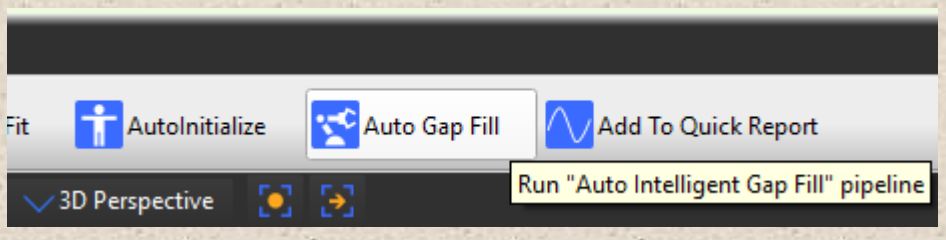
3. You can check the quality of captured data.



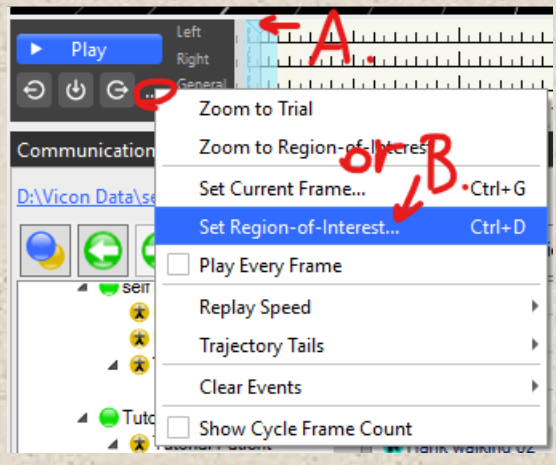
Commonly there are some gaps at the beginning and end of trial.

To eliminate gaps to zero, you can either

(1) Fill the Gap, OR



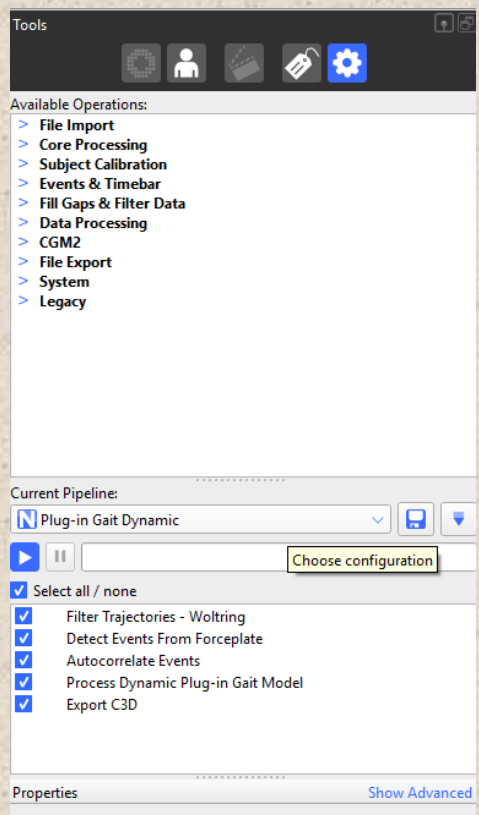
(2) choose the start/stop frame



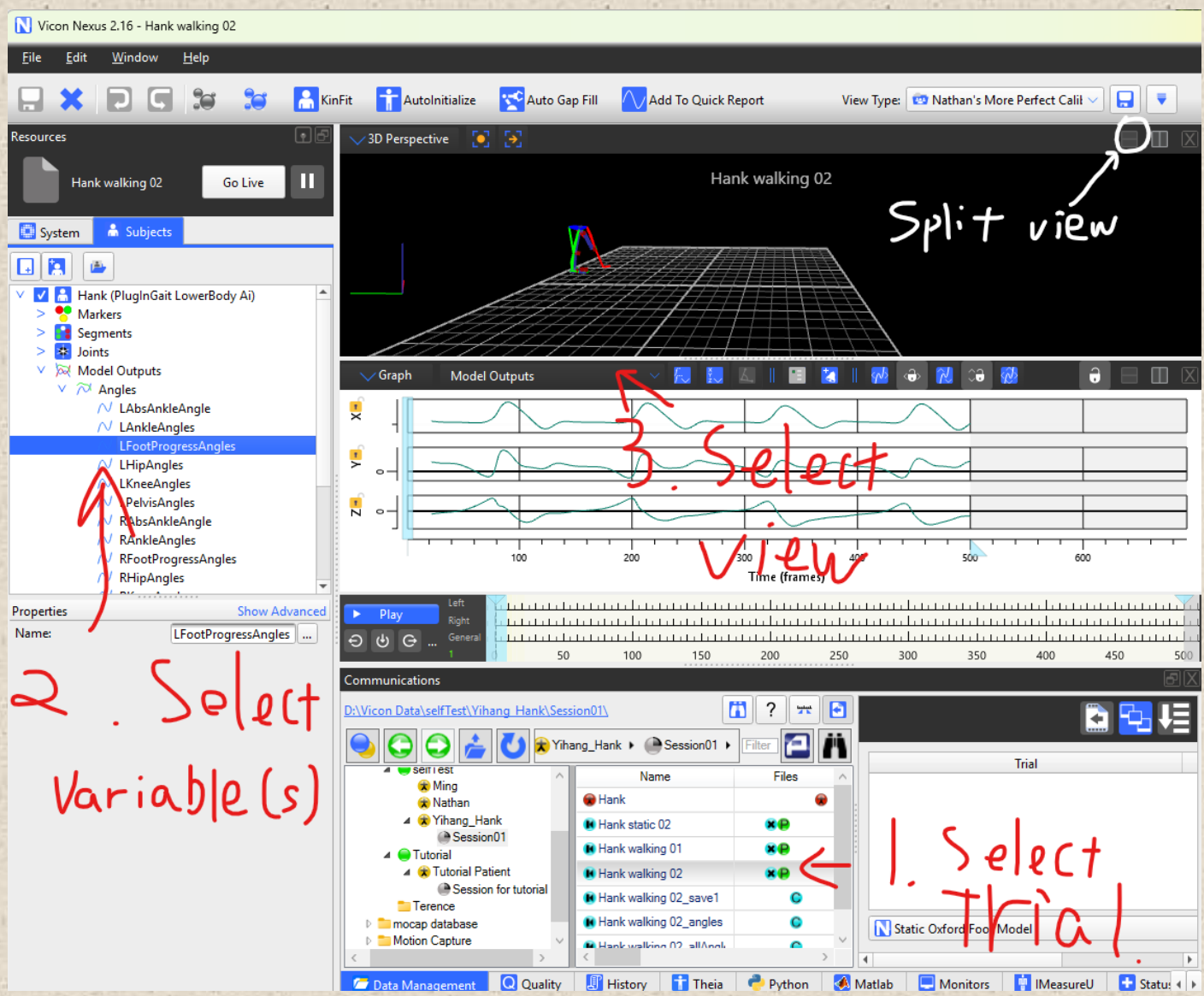
A. Drag the time axis on both sides, OR

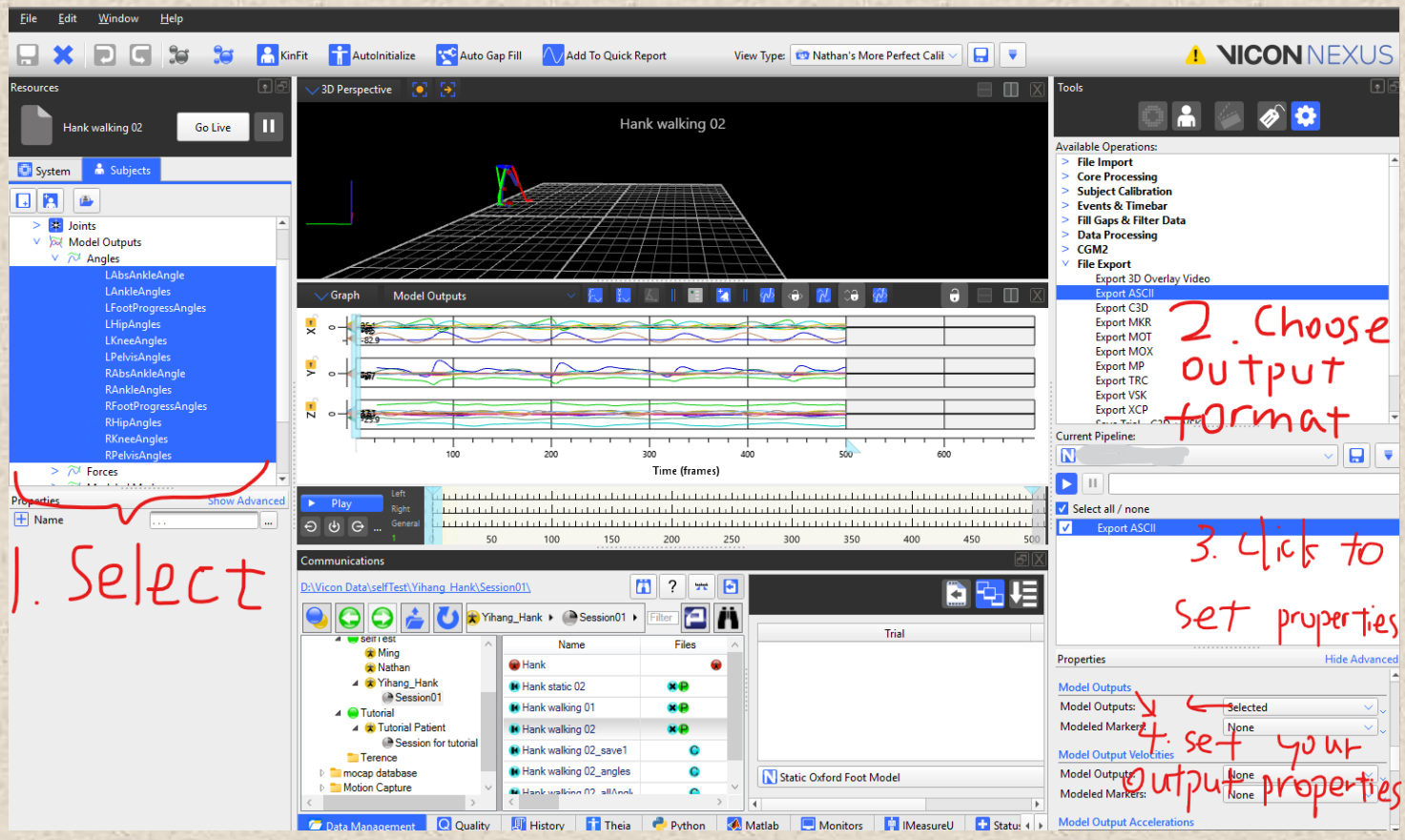
B. click "..." click "Set Region-of-interest"

4. Run the Plug-in Gait Dynamic pipeline.



3. VIEW AND EXPORT




File Edit Window Help
KinFit
AutoInitialize
Auto Gap Fill
Add To Quick Report
View Type: Nathan's More Perfect Calib
VICON NEXUS
Resources
Hank walking 02
Go Live
System
Subjects
Joints
Model Outputs
Angles
LAbsAnkleAngle
LAnkleAngles
LFootProgressAngles
LHipAngles
LKneeAngles
LPelvisAngles
RAbsAnkleAngle
RAnkleAngles
RFootProgressAngles
RHipAngles
RKneeAngles
RPelvisAngles
Forces
Properties
Show Advanced
Name
1. Select
3D Perspective
Hank walking 02
Graph
Model Outputs
Time (frames)
Play
Communications
D:\Vicon Data\selfTest\Yihang_Hank\Session01\
Yihang_Hank
Session01
Filter
Ming
Nathan
Yihang_Hank
Session01
Tutorial
Tutorial Patient
Session for tutorial
Terence
mocap database
Motion Capture
Name
Files
Hank
Hank static 02
Hank walking 01
Hank walking 02
Hank walking 02_save1
Hank walking 02_angles
Trial
Static Oxford Foot Model
Data Management
Quality
History
Theia
Python
Matlab
Monitors
IMeasureU
Status
Tools
Available Operations:
File Import
Core Processing
Subject Calibration
Events & Timebar
Fill Gaps & Filter Data
Data Processing
CGM2
File Export
Export 3D Overlay Video
Export ASCII
Export C3D
Export MKR
Export MOT
Export MOX
Export MP
Export TRC
Export VSK
Export XCP
2. Choose output format
Current Pipeline:
Select all / none
Export ASCII
3. Click to set properties
Properties
Hide Advanced
Model Outputs
Model Outputs:
Selected
Modeled Markers:
None
Model Output Velocities
Model Outputs:
None
Modeled Markers:
None
4. set your output properties
Model Output Accelerations

End of Motion Capture System User Guide